

Requirements Gathering for VR Simulators for Training: Lessons Learned for Globally Dispersed Teams

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ABSTRACT

We report an empirical study on the use of current VR technologies for requirements gathering in the field of simulation and training. We used synchronous and asynchronous traditional techniques plus collaborative virtual environments such as MozillaHubs and AltspaceVR. Our results show that requirements gathering in VR makes a difference in the process of requirements identification. We report advantages and shortcomings that can be useful for future practitioners. For example, we found that VR sessions allowed for better identification of dimensions and sizes. VR sessions for requirements gathering could also benefit from better pointers and better sound.

Index Terms: Human-centered computing—Collaborative and social computing—Empirical studies in collaborative and social computing; Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality

1 INTRODUCTION

Virtual Reality (VR) technologies could benefit a wide variety of application fields, given their availability and affordable prices. Although it has been reported that VR makes a difference in several industries and processes [1], novel commodity VR hardware opens new possibilities in fields that we have not studied in detail. One of such application areas is requirements elicitation, the identification of requirements between stakeholders and software developers. We study in this paper the application of VR technologies to this field, in particular in the development of VR simulators and training environments. The development process requires the understanding of the real space to be simulated. Besides, simulators require to reproduce real characteristics of a system; in particular, a physical 3D space, and VR could portray such characteristics.

At the beginning of any software development task, a team of software analysts, software developers, and end users, among other stakeholders, meet in order to define the requirements of their system. Such team may even be geographically distributed, a very common situation nowadays, specially due to current covid19 pandemic, so communication can also be diminished due to limitations such as the coordination of the communication channel [5]. VR could provide early prototypes of the simulation environment, similar to the reality that has to be simulated, and in this way provide an advantage over traditional elicitation techniques.

2 RELATED WORK

Bhimani [2] studied how VR could help in a requirements elicitation process in general. He concluded that VR could be beneficial for projects that involve physical spaces, where architectural, geographical, or visual issues should be considered. He also warned

about the extra costs and effort to develop the 3D scenario for these sessions. Therefore, we want to concentrate our efforts in simulators and trainers, a subset of the VR environments that can be created from physical environments. We also want to extend the discussion topics to contextual elements, since we believe that VR could also facilitate such discussion with stakeholders at early stages of development. Finally, we limit the cost and effort in the creation of the 3D scene for requirements elicitation (Section 3.1).

The challenges of collaboration in VR have been studied at different levels of detail and with different technologies [3]. There are also several techniques that have been used to facilitate remote collaboration in MR, such as live 3D reconstruction [8]. Communication cues in real life such as movements, hand gestures, people's appearance, and pronunciation features have been studied in order to identify shortcomings for asynchronous collaboration [3]. Understanding aids such as frames for areas of interest, rays to point certain features, and other awareness methods are reported in [6]. We were interested in what current tools have to offer for requirements gathering. Whence, these studies allowed us to select, in a more informed way, the tools for our study.

There have been studies on requirements elicitation for VR applications [9], as well as studies on the effectiveness of elicitation techniques in software engineering [5]. A missing point in these works is the remote collaboration among stakeholders, which is part of our focus and it is mentioned as important in [5].

3 EMPIRICAL STUDY

We performed an empirical study of the use of VR technologies in the requirements elicitation processes of two on-going projects in order to ascertain benefits and shortcomings of such technologies. On the one hand, we are developing a simulator of the control room of a ship in our Navy (P1 from now on) [4]. Our purpose is to facilitate training of both normal and emergency situations that a Ship Captain could encounter. A VR environment will facilitate such training anywhere, in situations that are difficult to reproduce in reality. On the other hand, we want to provide a training environment for physicians in which they can make decisions about the first minute of a baby's life after delivery [7] (P2 from now on). A VR environment will facilitate training in remote areas of our country.

In the requirement elicitation process, we planned in advance each session and the main questions that we wanted to be answered. It helped us to document and save all information from the meetings and identify key moments in them. All sessions were recorded and analyzed during this development process.

3.1 VR Based Requirements Gathering for VR Simulators and Trainers

In both projects we employed standard techniques for requirements gathering [5], interviews, document analysis and observation through videos of the current training methods. In P1, requirements gathering was a collaborative process between six Navy officials and two developers. P2, regularly involved two physicians, two developers and the project coordinator. Besides these current methods, we added two collaborative sessions with VR technologies to know how stakeholders reacted to the VR experiences, and how they could complement the traditional methodology for requirements

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elicitation. These sessions were used as soon as initial information was gathered and the environments were ready to visit.

Session 1 is designed as a proof of concept that VR technologies could be useful for the team and the domain at hand. It should show the context of the project and it should be easy to develop and easy to use for all stakeholders, especially for the non-technical ones. We opted for a shared experience in Mozilla Hubs based on 360 pictures that we created from panoramas taken from clients. Figure 1 shows the Delivery Room environment for Session 1 of P2. This session should focus on the possibilities and limitations of the technology and facilitate a discussion about the development agenda.

If the development team finds, from Session 1, that VR is useful for requirements gathering, Session 2 is designed to provide a more compelling VR experience. A 3D scene, similar to the 3D space required for the simulator, is created for the shared experience. Ideally, this environment should allow stakeholders to move freely and further discuss the expected functionality. Additionally, it should allow to validate the understanding of the available information and gather further useful data for the project. Figure 2 shows the environment for Session 2 of P2. We used Unity to create the 3D scene and AltSpaceVR to share it among stakeholders.

4 RESULTS

4.1 Main Lessons from Session 1

Session 1 was performed after several teleconferences with stakeholders, where the initial requirements and concepts about their domains were defined. The use of a scene in Mozilla Hubs created a cheaper and more efficient way to know the real locations. The virtual scenario provides a pointer which allowed stakeholders to direct the discussion to specific elements in the environment. This was useful to identify new requirements associated to elements and to find issues such as the appearance and final dimensions of the environment, and spatial distribution of objects. For example, *The trainer should be where "Person 1" is standing right now and he should do...; or if you turn around you should find "this object"*. Objects that should be in the simulator, objects that could be moved, missing objects, and non-relevant objects. However, even though participants used the pointer, sometimes it was hard to follow it (See the red pointer in Figure 1, for example), either because of occlusion with other participants or because of low contrast with the surroundings. Also, several times domain experts discussed about certain objects without properly pointing them. This made it difficult for other participants to understand the conversation, given the limitations to follow the elements they were referring to.

Even though we chose Mozilla Hubs due to its ease of use, it was nevertheless challenging for some non-technical people. Hence, we consider that a separate session for training is necessary. We also experienced some issues with the spatialized sound in Hubs, since not all participants could properly listen to a speaker in the environment. We decided to mute the voice in Hubs altogether, and use an alternative sound channel, i.e., Teams with standard sound, in order to assure that all stakeholders had the best audio possible. This session, in both projects, gave as a result the list of main objects in the simulation, and the overall dialog of each simulation.

4.2 Main Lessons from Session 2

Session 2 in both projects was designed to discuss important objects in the simulation, issues about the 3D models we wanted to have and their behavior in the final product. Stakeholders found the environment very compelling and they understood the differences between the mock-up in AltVR and the expected final product. Participants vividly discussed objects, their importance in the simulation, their relative sizes and their location within the 3D scene. There was a more compelling way to move around the 3D scenes, and stakeholders took advantage of that feature in order to explain in detail elements in the simulation.

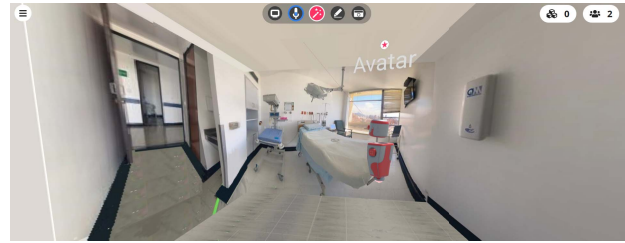


Figure 1: Environment for Session 1 in the Baby Delivery Project



Figure 2: Environment for Session 2 in the Baby Delivery Project

5 CONCLUSIONS AND FUTURE WORK

We present our experience in using VR technologies in the requirement elicitation process of VR simulators and trainers. VR facilitated the study of contextual issues at early stages of the development, and allowed stakeholders to identify issues related to position and scale of objects in the simulation. This experience could also benefit in the future from novel tools to identify the focus of a stakeholder, and ways to further facilitate the setup for a shared experience. In the future, we plan to incorporate these tools in our new developments, and identify ways to overcome the limitations of the environments we use for the shared experiences.

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